



INTRODUCTION TO CYBER FORENSICS



Version: 1.3a

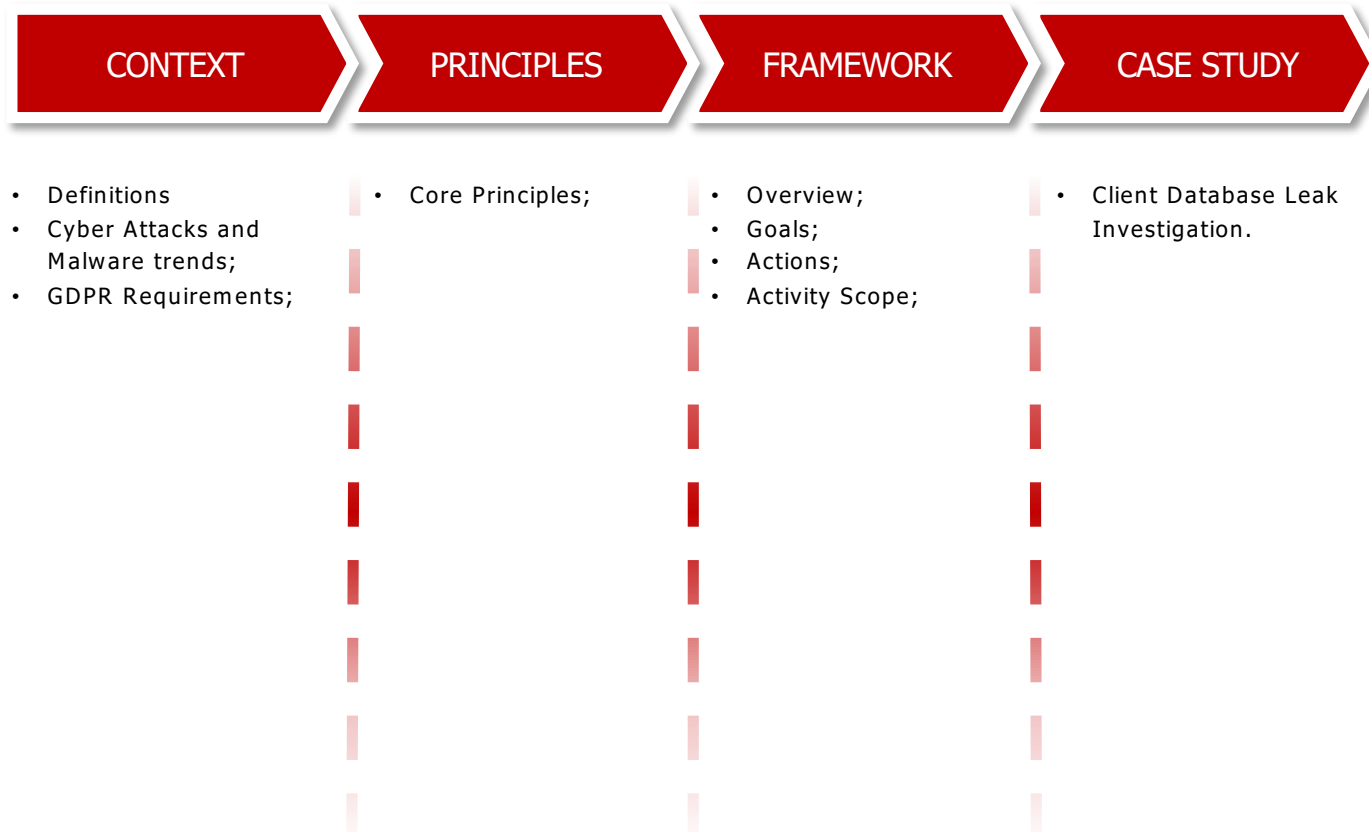
Date: 04/07/2018

Author: Sylvain Martinez

Reference: ESC6-MUSCL

Classification: Public

CONTENTS



FORENSIC SCIENCE

THE APPLICATION OF SCIENCE TO CRIMINAL AND CIVIL LAWS, DURING CRIMINAL INVESTIGATION, AS GOVERNED BY THE LEGAL STANDARDS OF ADMISSIBLE EVIDENCE AND CRIMINAL PROCEDURE.

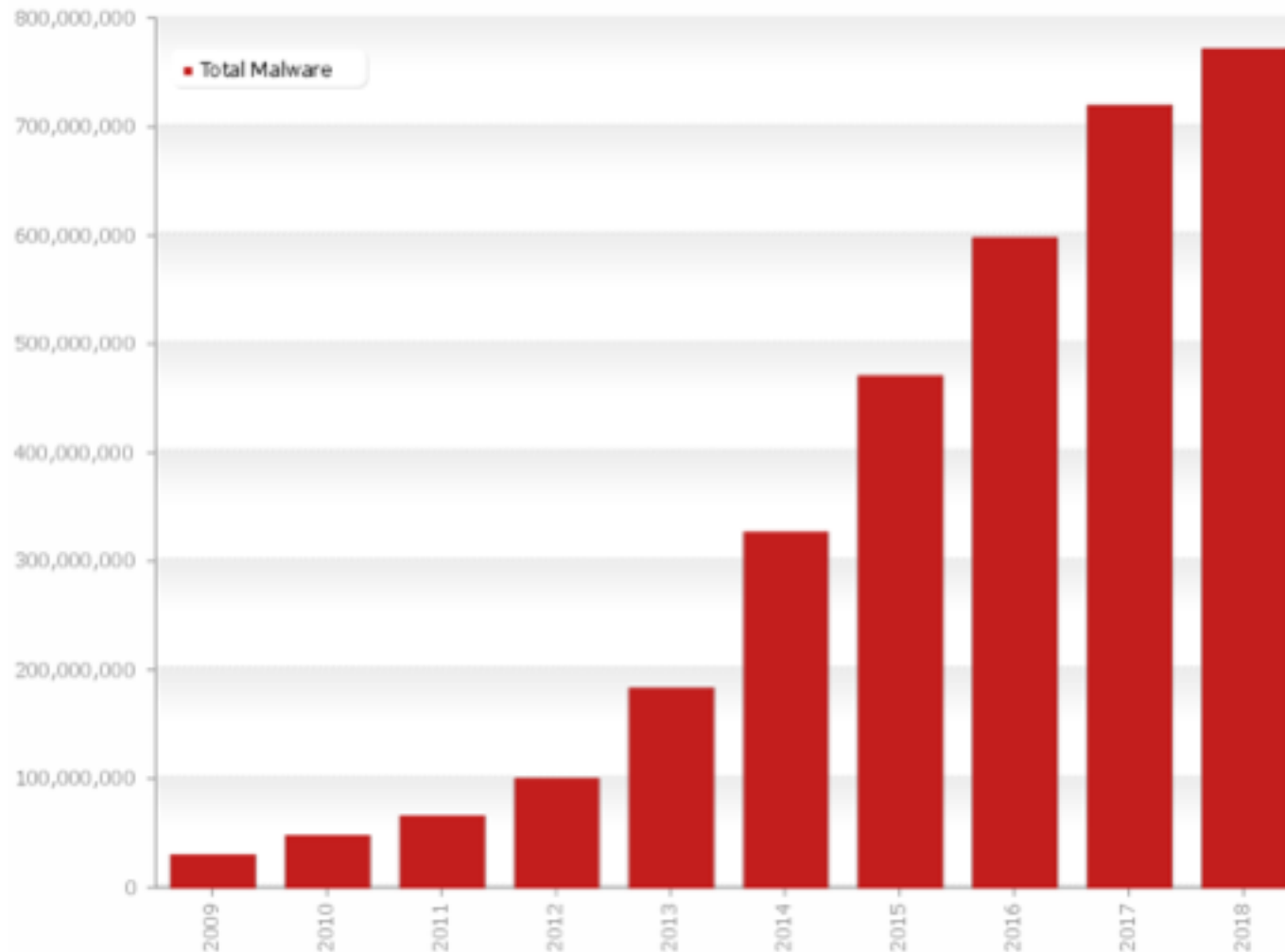
DIGITAL FORENSICS

A BRANCH OF FORENSIC SCIENCE ENCOMPASSING THE RECOVERY AND INVESTIGATION OF MATERIAL FOUND IN DIGITAL DEVICES, OFTEN IN RELATION TO COMPUTER CRIME

CYBER/COMPUTER FORENSICS

A BRANCH OF DIGITAL FORENSIC SCIENCE, THE APPLICATION OF **INVESTIGATION AND ANALYSIS TECHNIQUES** TO **GATHER AND PRESERVE EVIDENCE** FROM A PARTICULAR **COMPUTING DEVICE** IN A WAY THAT IS SUITABLE FOR PRESENTATION IN A **COURT OF LAW**

CYBER ATTACKS AND MALWARE TRENDS



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**ARTICLE 33
72H REPORTING**

NATURE OF THE BREACH?
(*WHO? WHERE? HOW?*)

POTENTIAL IMPACT?

WHAT HAS BEEN DONE TO PREVENT THE BREACH?
(*CONTROLS? PROCESSES?*)

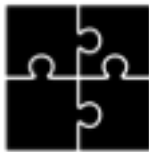
CORE PRINCIPLES



NEVER FORGET THE « S »!!



CHAIN OF CUSTODY



PRESERVATION OF INTEGRITY



ONLINE/OFFLINE ?



ACTIVITY GOALS

GOALS

IDENTIFICATION

Impact/Target/Technique

COLLECTION

Evidence of compromise

ATTRIBUTION

Source of Attack

**ACTIVITY
SCOPE**

CONTEXT

LOGS

FILESYSTEM

CONFIG

NETWORK

MEMORY

ADVANCED

ACTIONS

ACQUISITION

ANALYSIS

REPORTING

IDENTIFICATION

Impact/Target/Technique

COLLECTION

Evidence of compromise

ATTRIBUTION

Source of Attack

WHO?

WAS TARGETED?

WAS RESPONSIBLE?

WHAT?

WAS COMPROMISED?

WAS STOLEN/MODIFIED?

WHERE?

THE CONTROLS FAILED?

THE DATA WENT?

HOW?

THEY HACKED?

WAS IT STOPPED?

WHY?

THEY TARGETED YOU?

WAS IT SUCCESSFUL?

ACQUISITION

ANALYSIS

REPORTING

IDENTIFICATION OF EVIDENCE

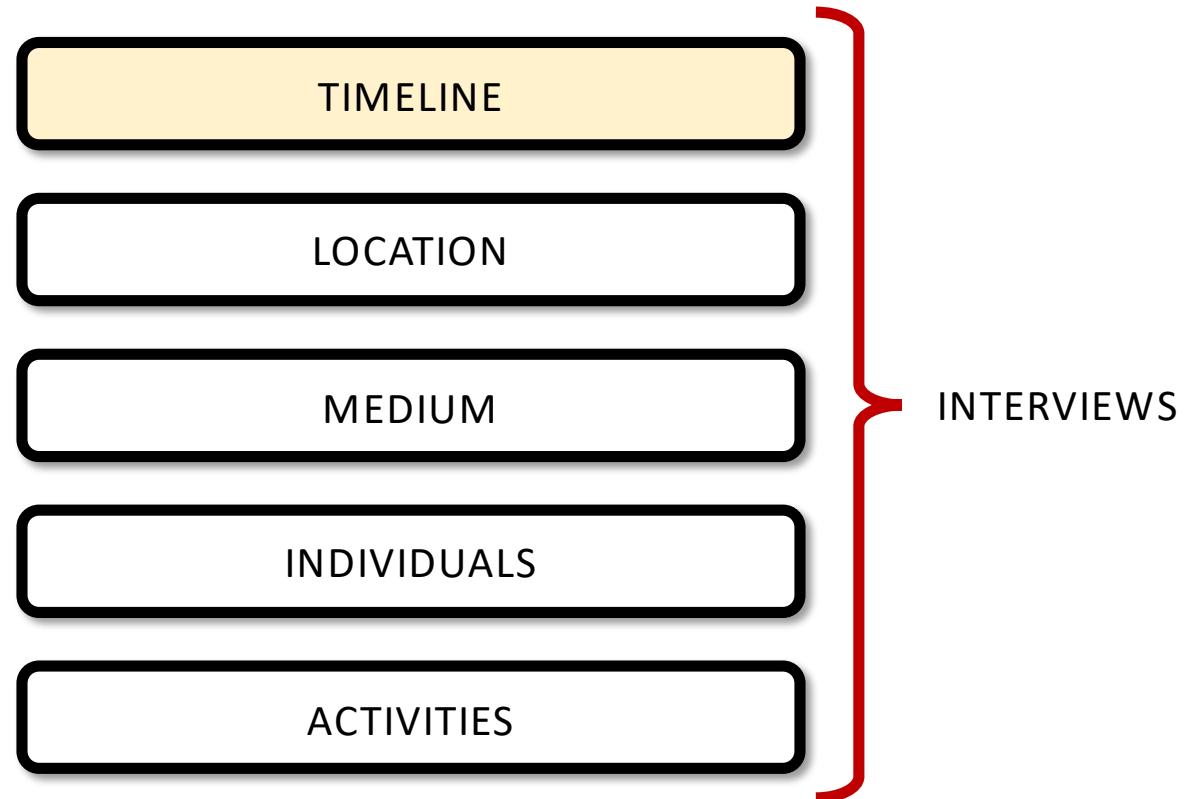
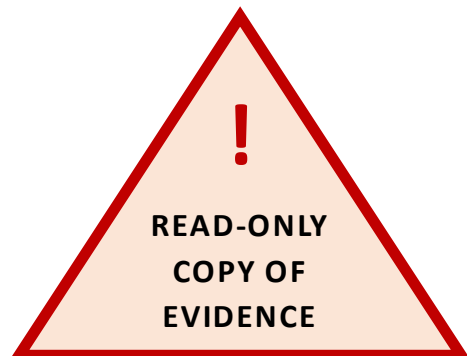
PRESERVATION OF EVIDENCE

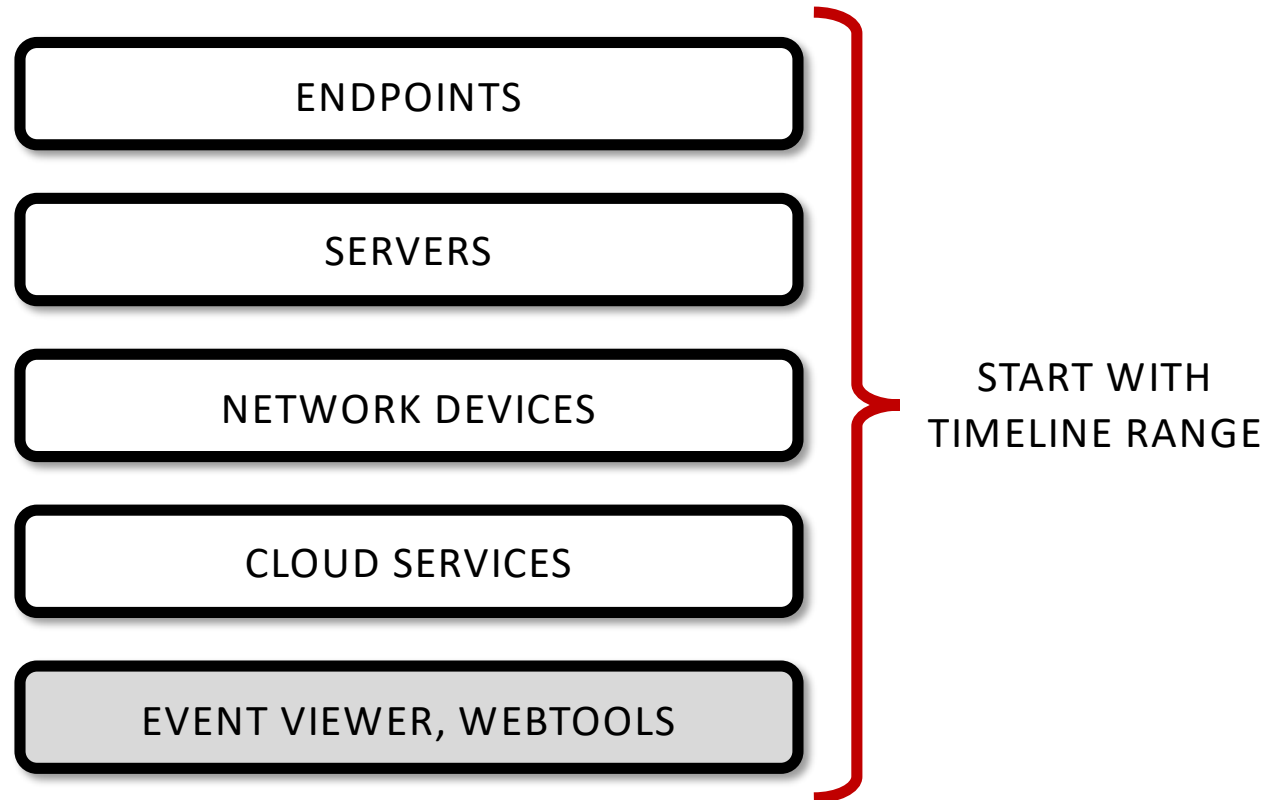
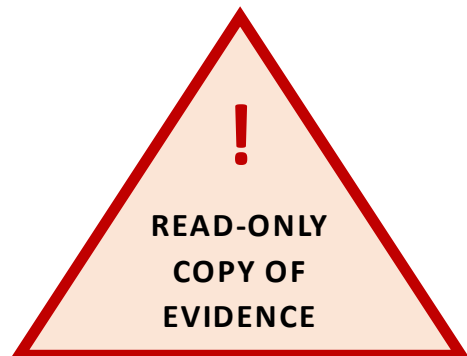
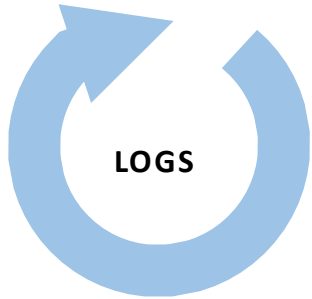
COLLECTION OF EVIDENCE

ANALYSIS OF EVIDENCE

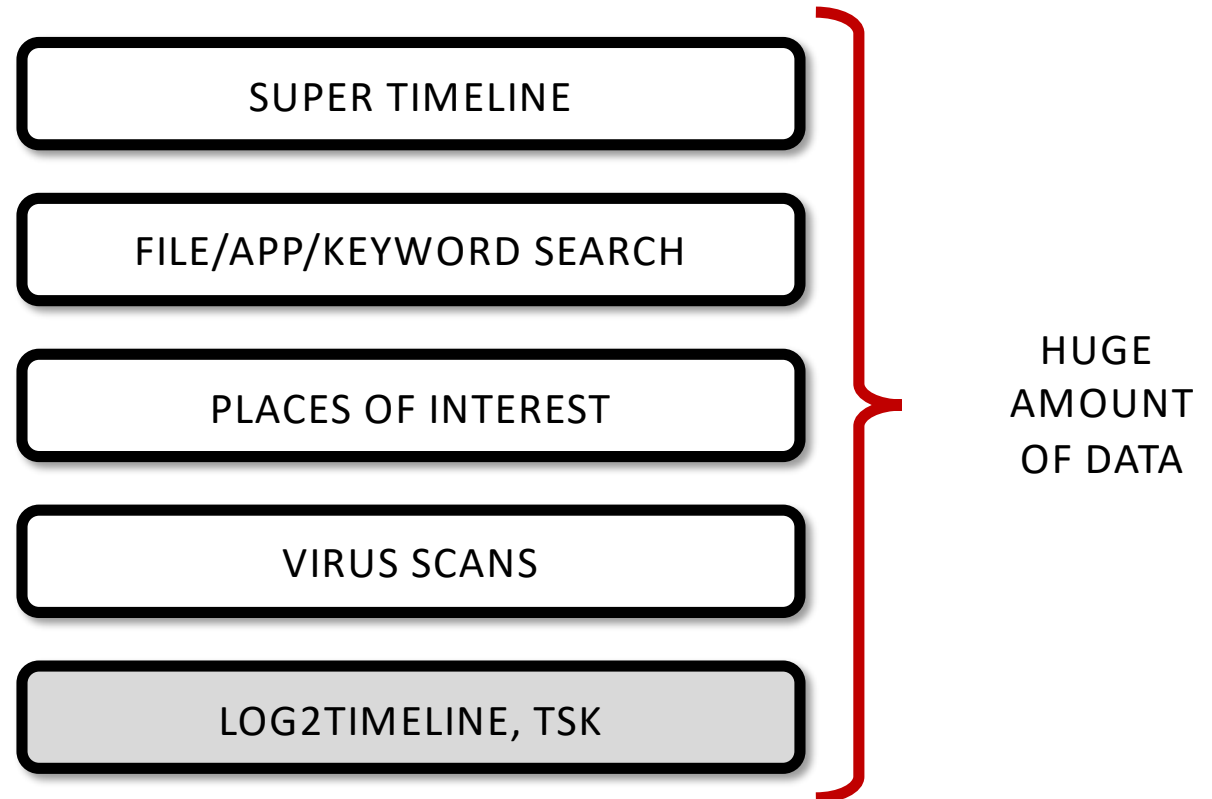
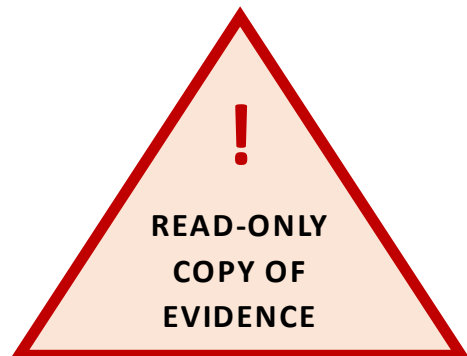
DOCUMENTATION OF EVIDENCE

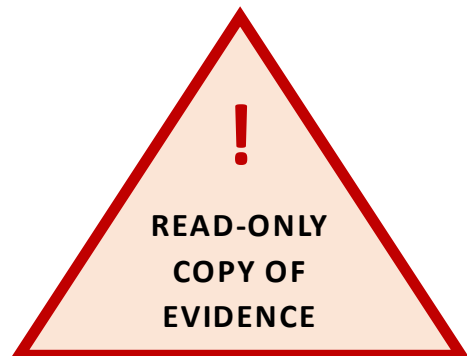
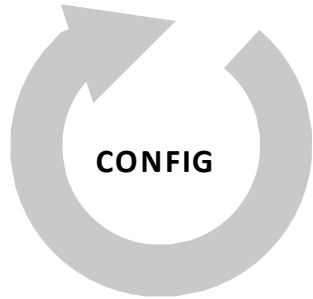
PRESENTATION OF EVIDENCE





ACTIVITY - FILESYSTEM





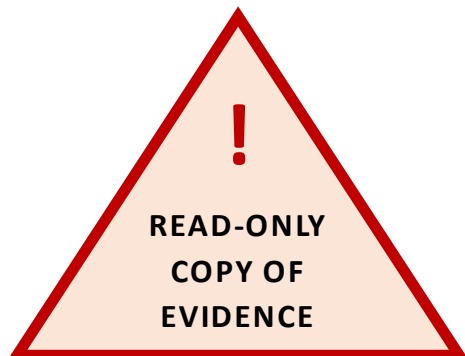
REGISTRY KEY HIVE

SYSTEM FILES

APPLICATION CONFIGURATION

RECENT CHANGES/INSTALLATIONS

REGEDIT/HIJACKTHIS/GREP



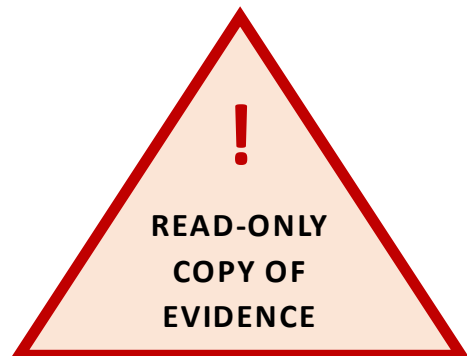
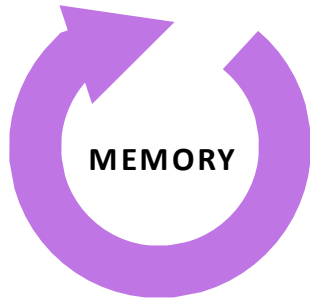
SOURCE / DESTINATION ACTIVITIES

PROTOCOL USED

TRAFFIC CONTENT ANALYSIS

IDS ANALYSIS

WIRESHARK / TCPDUMP



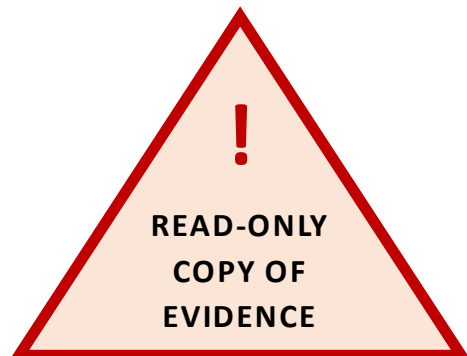
DUMP MEMORY / PAGE FILES

RUNNING PROCESSES

BINARY INSPECTION

HIDDEN DATA

VOLATILITY / REKAL



USER ACTIVITY SIMULATION

MALWARE REVERSE ENGINEERING

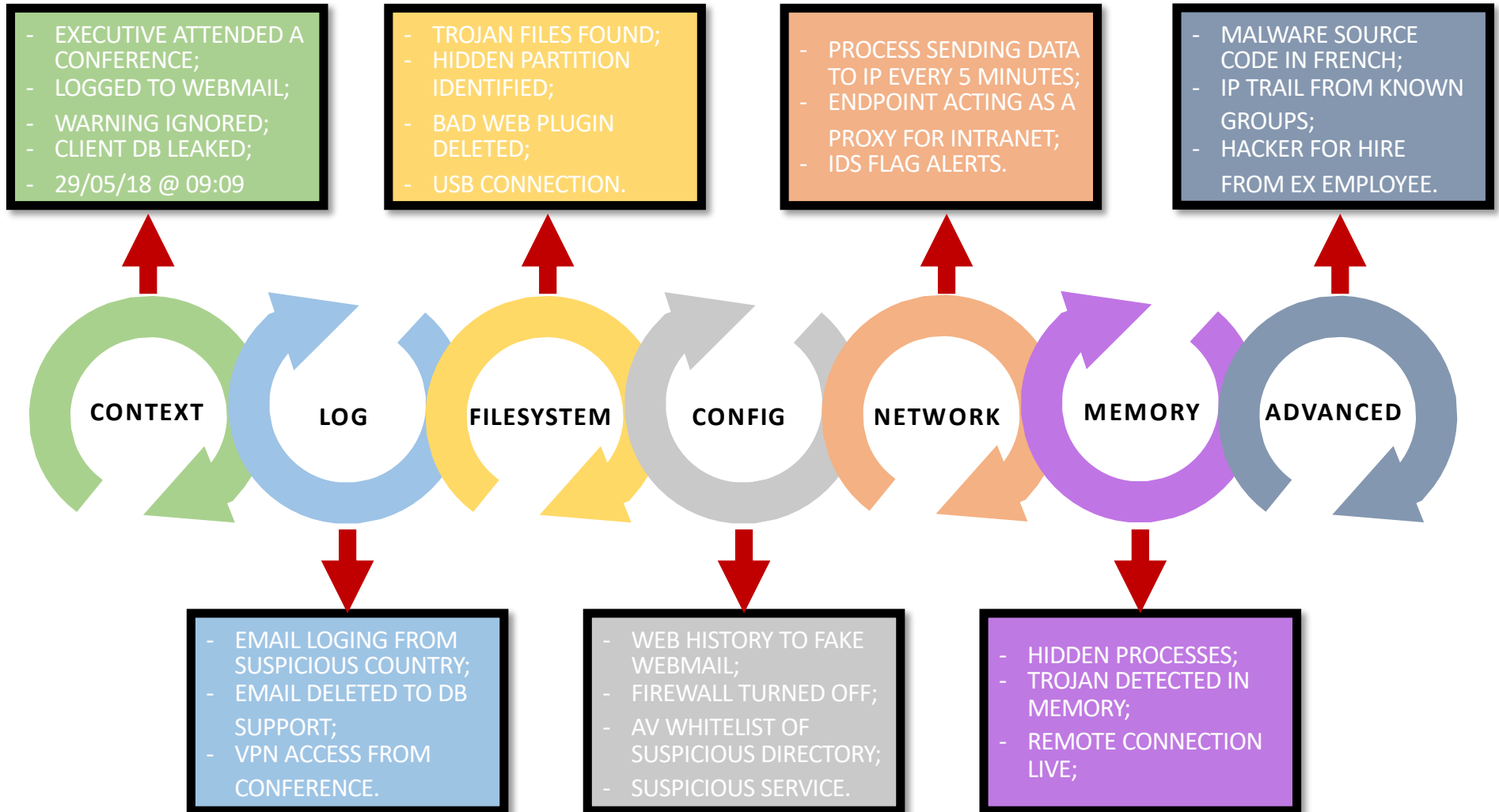
MALWARE SANDBOXING

HONEYPOTS

HACKER COMMUNICATION

DANGEROUS!

CLIENT DATABASE LEAK INVESTIGATION



ABOUT ELYSIUMSECURITY LTD.

ElysiumSecurity provides a portfolio of **Strategic** and **Tactical Services** to help companies protect and respond against Cyber Security Threats. We differentiate ourselves by offering discreet, tailored and specialized engagements.

Operating in **Mauritius** and in the **United Kingdom**, our *boutique style* approach means we can easily adapt to your business operational model and requirements to provide a personalized service that fits your working environment.

ElysiumSecurity provides **high level expertise** gathered through years of **best practices experience** in large international companies allowing us to provide advice best suited to your **business operational model** and **priorities**.

ElysiumSecurity provides **practical expertise** to **identify vulnerabilities**, assess their **risks** and **impact**, **remediate** those risks, **prepare** and **respond** to incidents as well as raise **security awareness** through an organization.

